CS 237: PROBABILITY IN COMPUTING

INSTRUCTORS: TIAGO JANUARIO, NATHAN KLEIN

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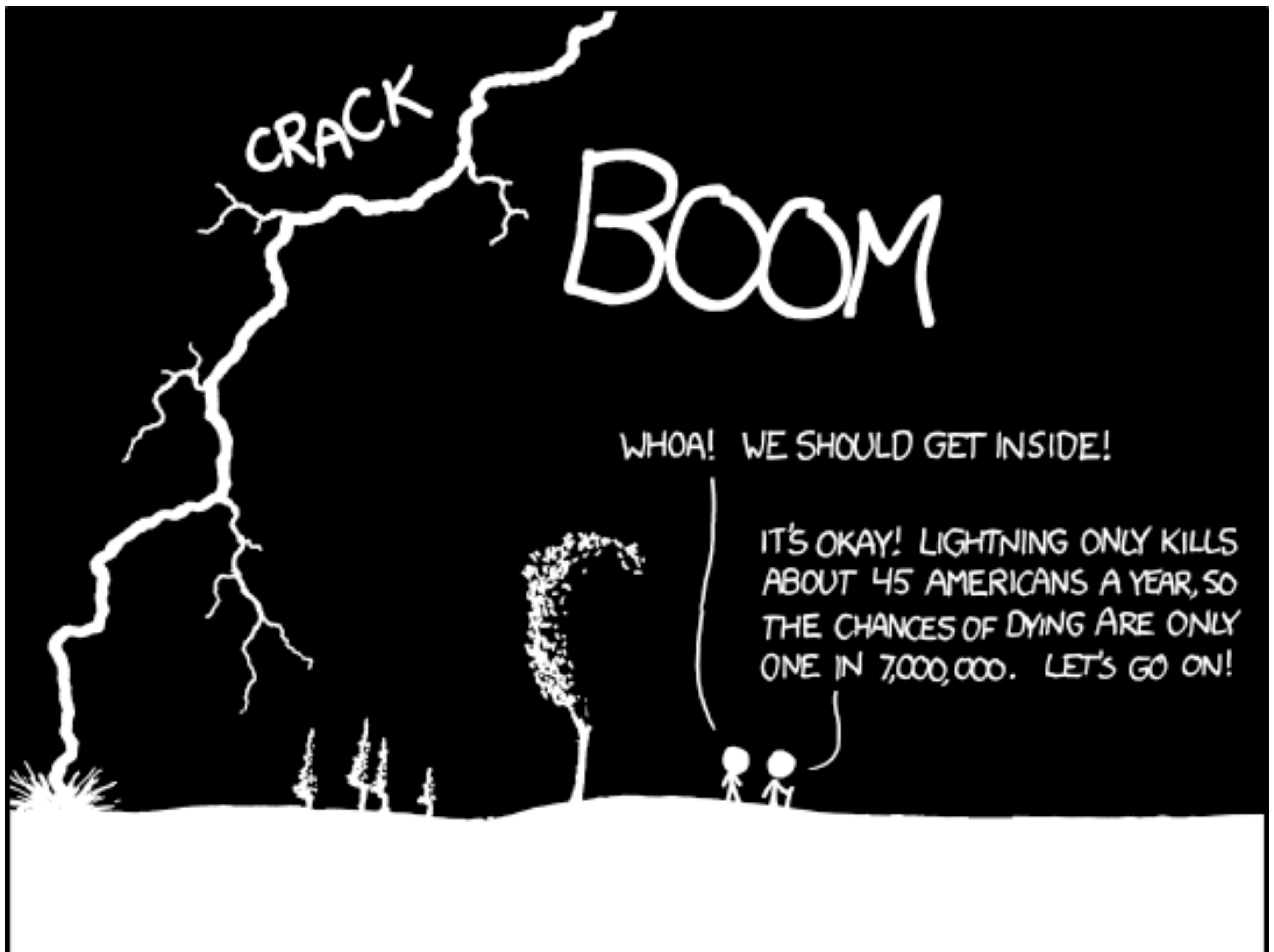

THE ANNUAL DEATH RATE AMONG PEOPLE WHO KNOW THAT STATISTIC IS ONE IN SIX.

LECTURE 7

Last time
- Random Variables
- Discrete Probability Distributions

Today
- Continuous Probability Distributions
- Probability Distributions Properties

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RANDOM VARIABLES

- Definition: A random variable X is a function whose domain is the sample space and co-domain is a subset of $\mathbb{R}$:

$$X: \ \Omega \to \mathbb{R}$$

- X maps each outcome $\omega \in \Omega$ to a value to a value $X(\omega) \in \mathbb{R}$:

Note: The name "random variable" is a misnomer, random variables are actually functions

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DISCRETE DISTRIBUTION FUNCTIONS

outputs probabilities for each value k in the range of your random variable

- Let X be a discrete random variable
- The probability mass function (PMF) of X is the following function:

$$f_X: \mathbb{R} \to [0,1], f_X(k) = Pr(X = k) \text{ for all } k \in \mathbb{R}$$

$K \in Range(X)$

- The cumulative density function (CDF) of X is the following function:

$$F_X: \mathbb{R} \to [0,1], F_X(k) = \underbrace{Pr(X \leq k)}_{\text{sum of probabilities}} \text{ for all } k \in \mathbb{R}$$

$$F_X(k) = Pr(X \leq k) = \sum_{k \leq x,\ \forall x \in Range(X)} Pr(X = k)$$

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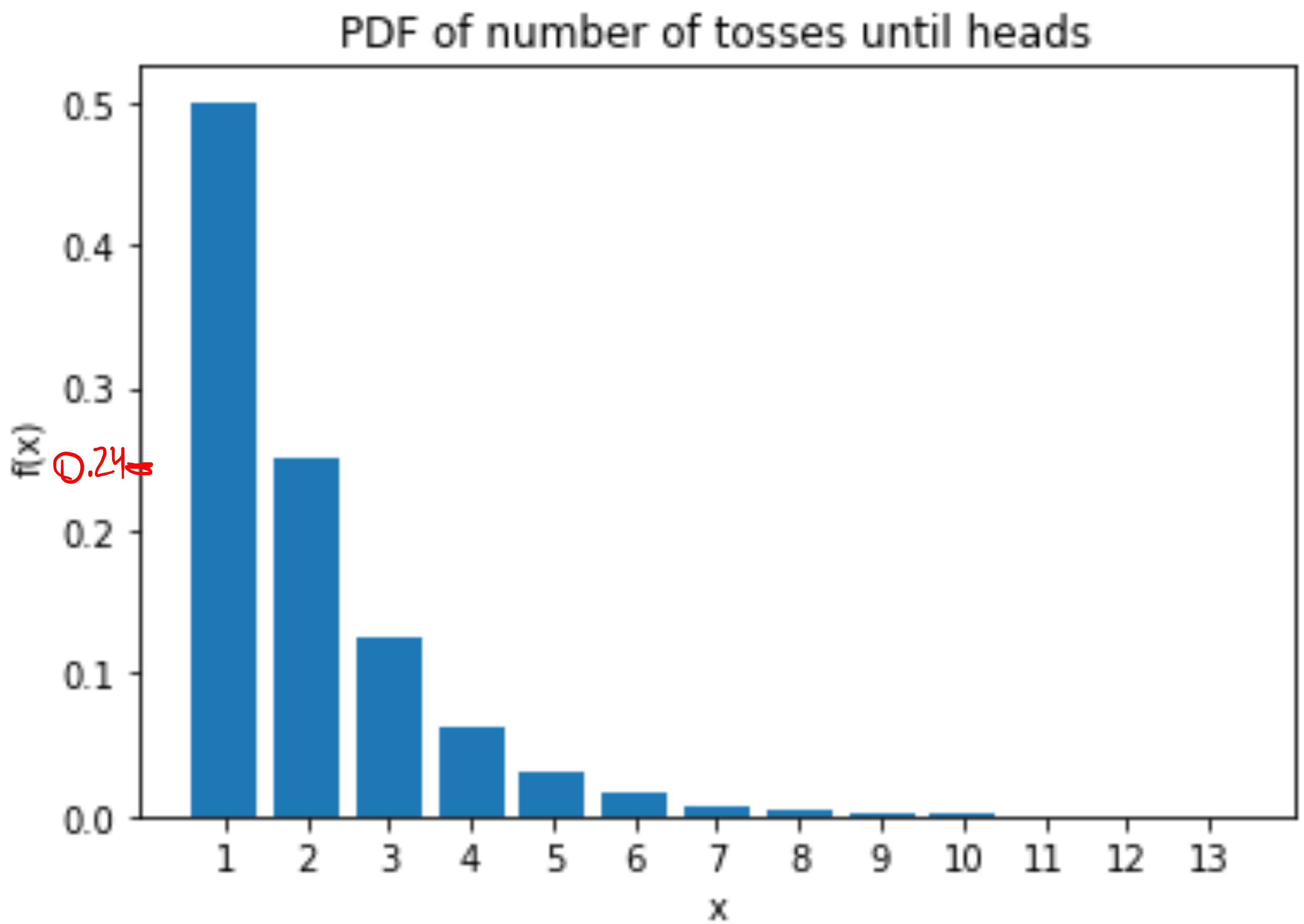
DISTRIBUTION FUNCTIONS

Range(C) = {1, 2, 3, 4, ... }

- Random variable: number of tosses until Heads



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DISTRIBUTION FUNCTIONS

- Random variable: number of tosses until Heads

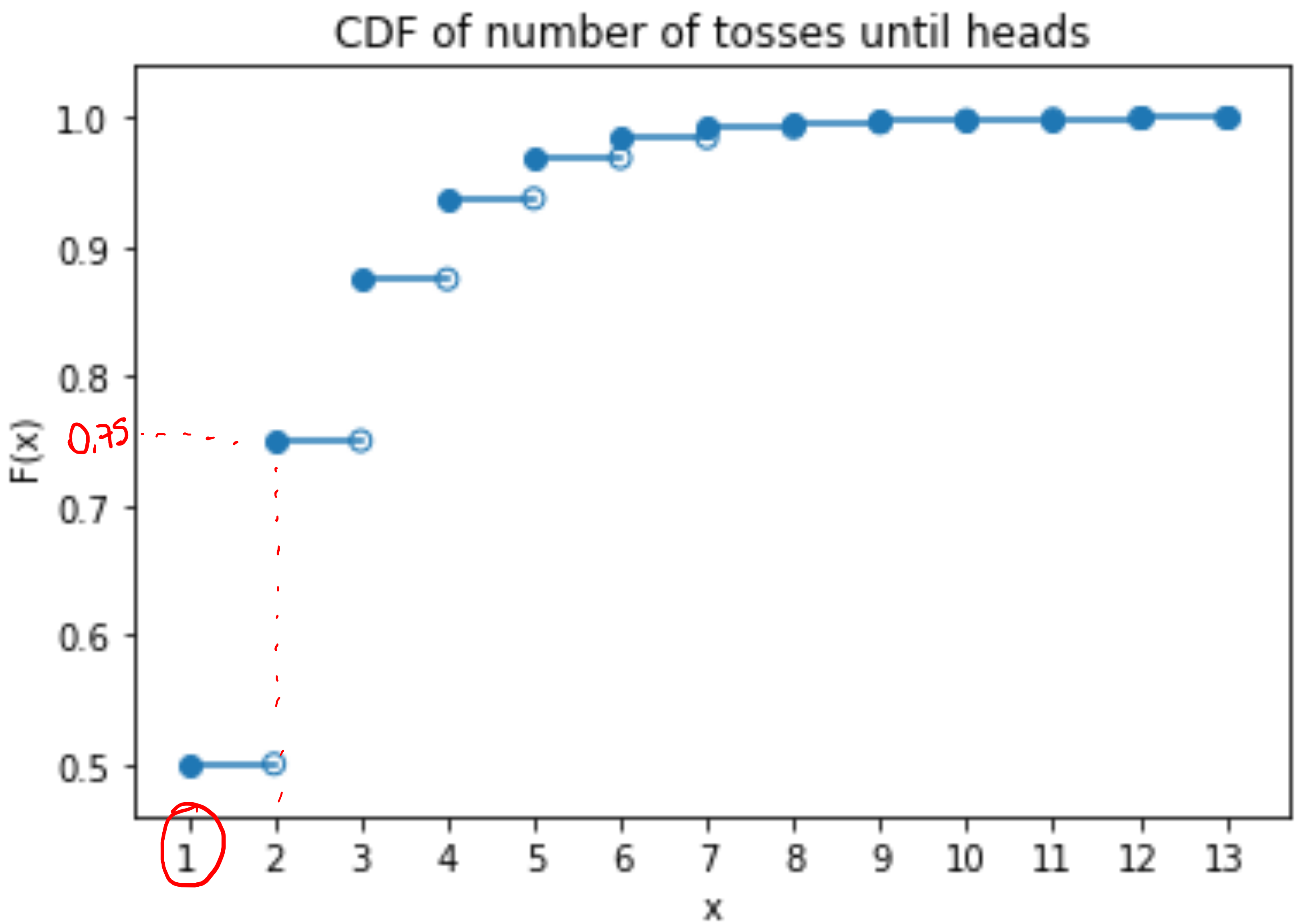
$Pr(X \le 1) = Pr(X = 1) = 0.5$

$Pr(X \le 2) = 0.75$

$= Pr(X = 1 \cup X = 2)$

$= Pr(X = 1) + Pr(X = 2)$

$= 0.5 + 0.25$



$|\Omega| = 36$

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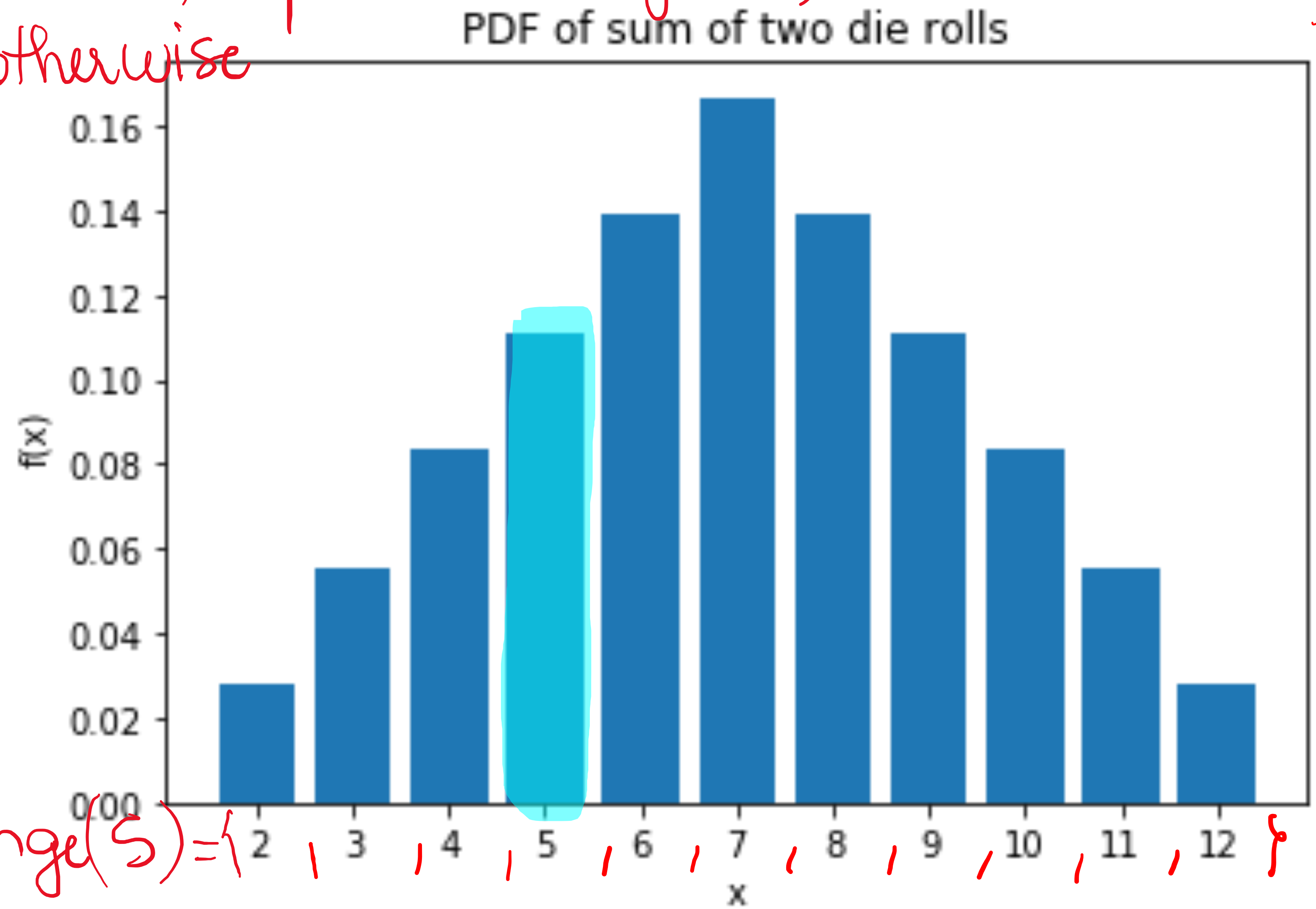
DISTRIBUTION FUNCTIONS

- Random variable: sum of two rolls of a fair die S_7

$$f_X(k) = \begin{cases} \Pr(X = k), & \text{if } k \in \text{Range}(X) \\ 0, & \text{otherwise} \end{cases}$$

	1	2	3	4
1	2			5
2			5	
3		5		
4	5			
⋮				



$\text{range}(S) = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

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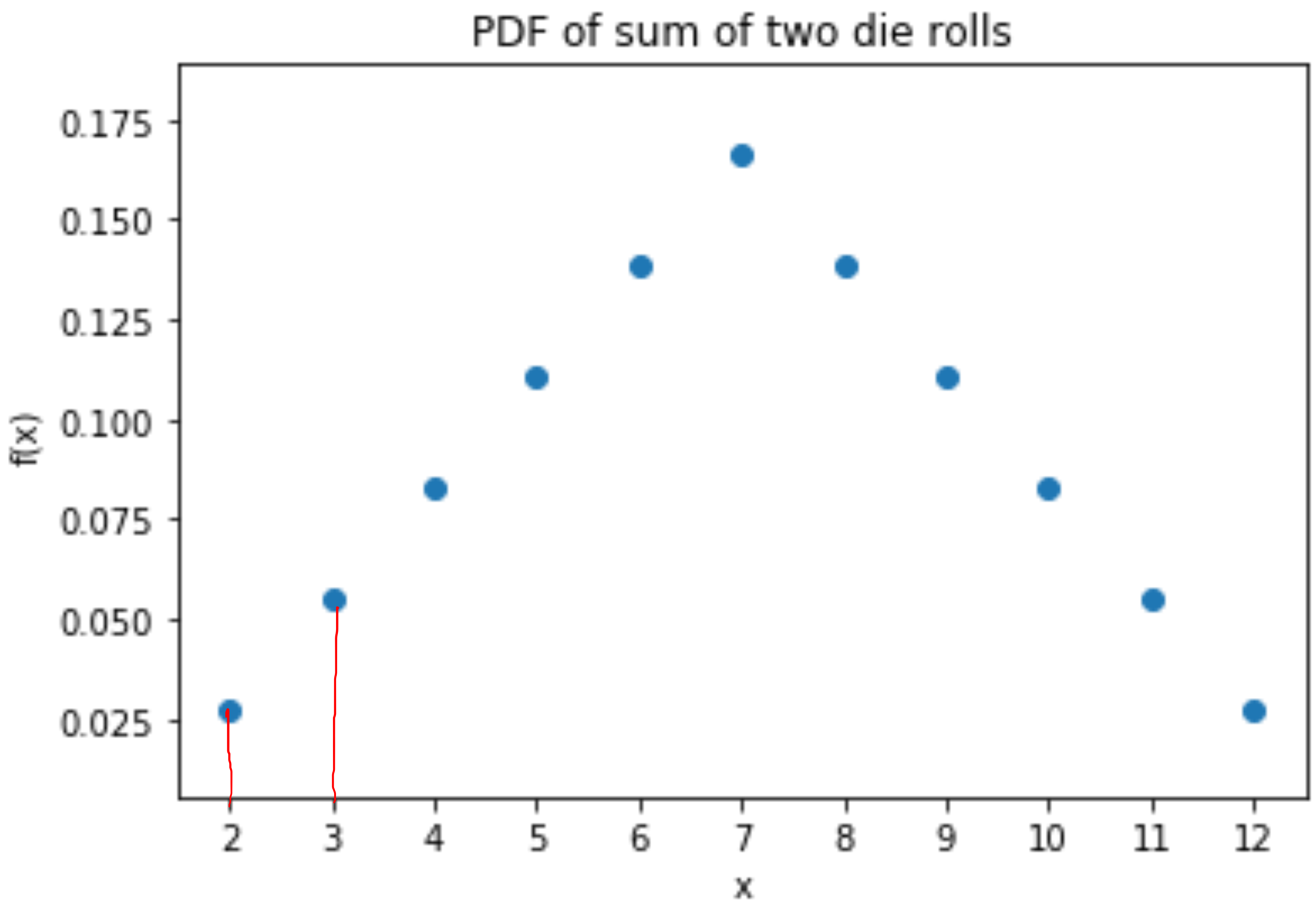
DISTRIBUTION FUNCTIONS

$Pr(X=2 \cup X=3) = Pr(X=2) + Pr(X=3)$

$Pr(X=3) = Pr(\{(1,2),(2,1)\})$

- Random variable: sum of two rolls of a fair die



DISTRIBUTION FUNCTIONS

?

$$\Pr(X=8) = F_X(8) - F_X(7)$$

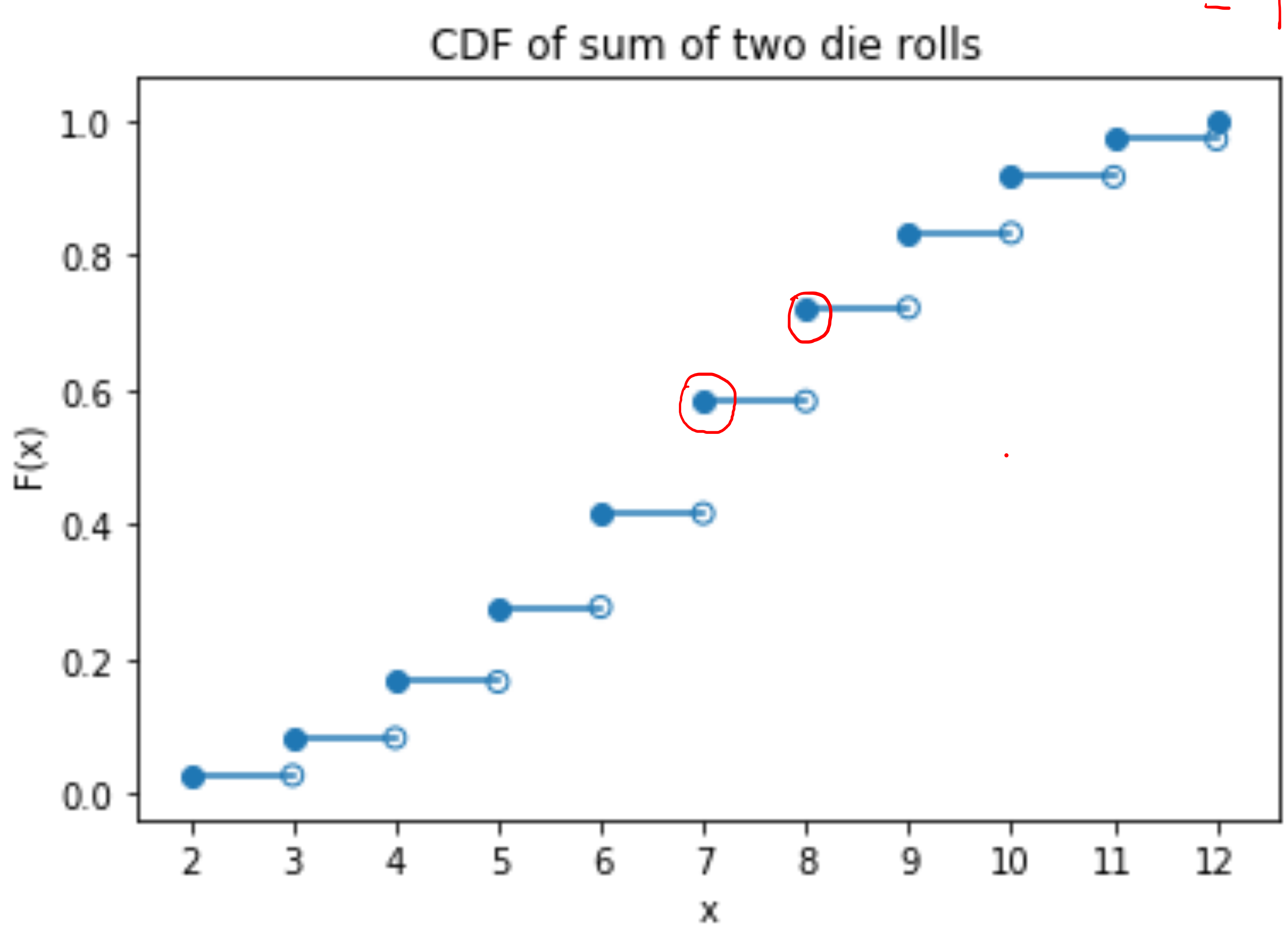
$$= \underbrace{\Pr(X \le 8)} - \Pr(X \le 7)$$

$$= \Pr(X=8) + \cancel{\Pr(X \le 7)}$$

$$- \cancel{\Pr(X \le 7)}$$

$$= \Pr(X=8)$$

- Random variable: sum of two rolls of a fair die



CDF <-> PMF

- On the whiteboard/blackboard, I went through the steps (a simple algorithm) to compute the CDF given the PMF of a discrete random variable as input.
- I also showed how to compute the PMF given the CDF of a discrete random variable.
- So, bottom line: PMFs and CDFs contain the same information and given one, you can compute the other.

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ASK THE AUDIENCE

X is a random variable

values of X	-1	4	7	10
CDF F(x)	0.5	0.75	0.8	1

What is $Pr(X \leq 4)$?

(A) 0.2 (B) 0.25 (C) 0.5 (D) 0.75

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ASK THE AUDIENCE

$$\Pr(X=4) = \boxed{F_X(4) - F_X(-1)}$$
$$= \Pr(X \le 4) - \Pr(X \le -1)$$
$$= \Pr(X=4) + \cancel{\Pr(X=-1)} - \cancel{\Pr(X=-1)}$$

X is a random variable $= \Pr(X=4)$

values of X	-1	4	7	10
CDF F(x)	0.5	0.75	0.8	1
$f(x)$	0.5	0.25	0.05	0.2

What is $\boxed{Pr(X = 4)?}$

(A) 0.2 $\boxed{\text{(B) 0.25}}$ (C) 0.5 (D) 0.75

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ASK THE AUDIENCE

X is a random variable

values of X	-1	4	7	10
CDF F(x)	0.5	0.75	0.8	1

What is $Pr(-1 < X \leq 4)$?

Pr(X=4)

(A) 0.2 (B) 0.25 (C) 0.5 (D) 0.75

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ASK THE AUDIENCE

X is a random variable

values of X	-1	4	7	10
CDF F(x)	0.5	0.75	0.8	1

What is $Pr(-1 \leq X \leq 4)$? $= Pr(X \leq 4)$

(A) 0.2 (B) 0.25 (C) 0.5 (D) 0.75

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DISTRIBUTION FUNCTIONS: EXAMPLE

- Alice pick a card uniformly at random from a standard deck
- Alice donates some money to charity as follows:
  - If the card is a spade (♠) or an Ace: \$5.50
  - If the card is a face card (J, Q, K) but not ♠: \$3.00
  - If the card is an even number (2,4,6, 8, 10) but not ♠: \$0.75
- Let X = amount Alice donates
- Find the PMF and CDF of X

f_X F_X

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DISTRIBUTION FUNCTIONS: EXAMPLE

- Let X = amount Alice donates

$$f_X(k) = \Pr(X=k) = \begin{cases} 12/52, & \text{if } k = \$0 \\ 15/52, & \text{if } k = \$0.75 \\ 9/52, & \text{if } k = \$3.00 \\ 16/52, & \text{if } k = \$5.50 \\ 0, & \text{otherwise} \end{cases}$$

DISTRIBUTION FUNCTIONS: EXAMPLE

- Let X = amount Alice donates

$$F_X(K) = \Pr(X \le K) = \begin{cases} 0, & \text{if } K < 0 \\ 12/52, & \text{if } 0 \le K < 0.75 \\ \frac{27}{52}, & \text{if } 0.75 \le K < 3 \\ \frac{36}{52}, & \text{if } 3 \le K < 5.5 \\ 1, & \text{if } K \ge 5.5 \end{cases}$$

DISTRIBUTION FUNCTIONS: MOTIVATION

- The PMF and CDF do not involve the sample space
- There are many random variables arising in different probability spaces with the same PMF/CDF
- Random variables with the same PMF/CDF behave very similarly
- Can classify random variables into distributions and study them in a unified way

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DISTRIBUTION FUNCTIONS: CONTINUOUS

computes the area under the curve given f(x) for the event E = [a, b]

- Let X be a continuous random variable
- The probability density function (PDF) of X is a function f satisfying

$$f(x) \geq 0, Pr(X \in [a,b]) = \boxed{\int_a^b f(x)dx}$$

- The cumulative density function (CDF) of X is the following function:

$$F: \mathbb{R} \to [0,1], F(x) = Pr(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

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PROPERTIES OF THE PDF (DISCRETE AND CONTINUOUS)

- The PDF is non-negative and normalized (total Pr is 1):

$$f(x) \geq 0 \text{ for all } x \in \mathbb{R}$$

no value is less than zero

$$\int_{-\infty}^{\infty} f(x)dx = \Pr(-\infty < X < +\infty) = 1$$

no probability value is greater than 1

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DISTRIBUTION FUNCTIONS (CONTINUOUS)

- Let X be a continuous random variable
- How should we define the PDF and CDF?

Recall the CDF for discrete random variables:

$$F: \mathbb{R} \to [0,1], F(x) = Pr(X \leq x) \text{ for all } x \in \mathbb{R}$$

- This still makes sense for continuous random variables
- **Example:** X is uniform over [0,1]

$$F_X(x) = Pr(X \leq 1/3) = Pr(X \in [0,1/3]) = 1/3$$

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DISTRIBUTION FUNCTIONS (CONTINUOUS)

- Let X be a continuous random variable
- How should we define the PDF and CDF?

Recall the PMF for discrete random variables:

$f: \mathbb{R} \to [0,1], \ \ f(x) = \Pr(X = x)$ for all $x \in \mathbb{R}$

This is no longer informative for continuous random variables:

$$\Pr(X = x) = 0 \quad \text{for all } x \in \mathbb{R}$$

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DISTRIBUTION FUNCTIONS (CONTINUOUS)

- Let X be a continuous random variable
- How should we define the PDF and CDF?

Recall the PMF for discrete random variables:

$$f\colon \mathbb{R} \to [0,1],\ \ f(x) = \Pr(X = x) \quad \text{for all } x \in \mathbb{R}$$

For continuous rvs, we use the derivative of the CDF F:

$$f(x) = F'(x) \quad \text{for all } x \in \mathbb{R}$$

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DISTRIBUTION FUNCTIONS (CONTINUOUS)

- Let X be a continuous random variable
- How should we define the PDF and CDF?

For continuous rvs, we use the derivative of the CDF F:

$$f(x) = F'(x) \quad \text{for all } x \in \mathbb{R}$$

Example: X is uniform over $[0,1]$

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

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DISTRIBUTION FUNCTIONS (CONTINUOUS)

- Let X be a continuous random variable
- How should we define the PDF and CDF?

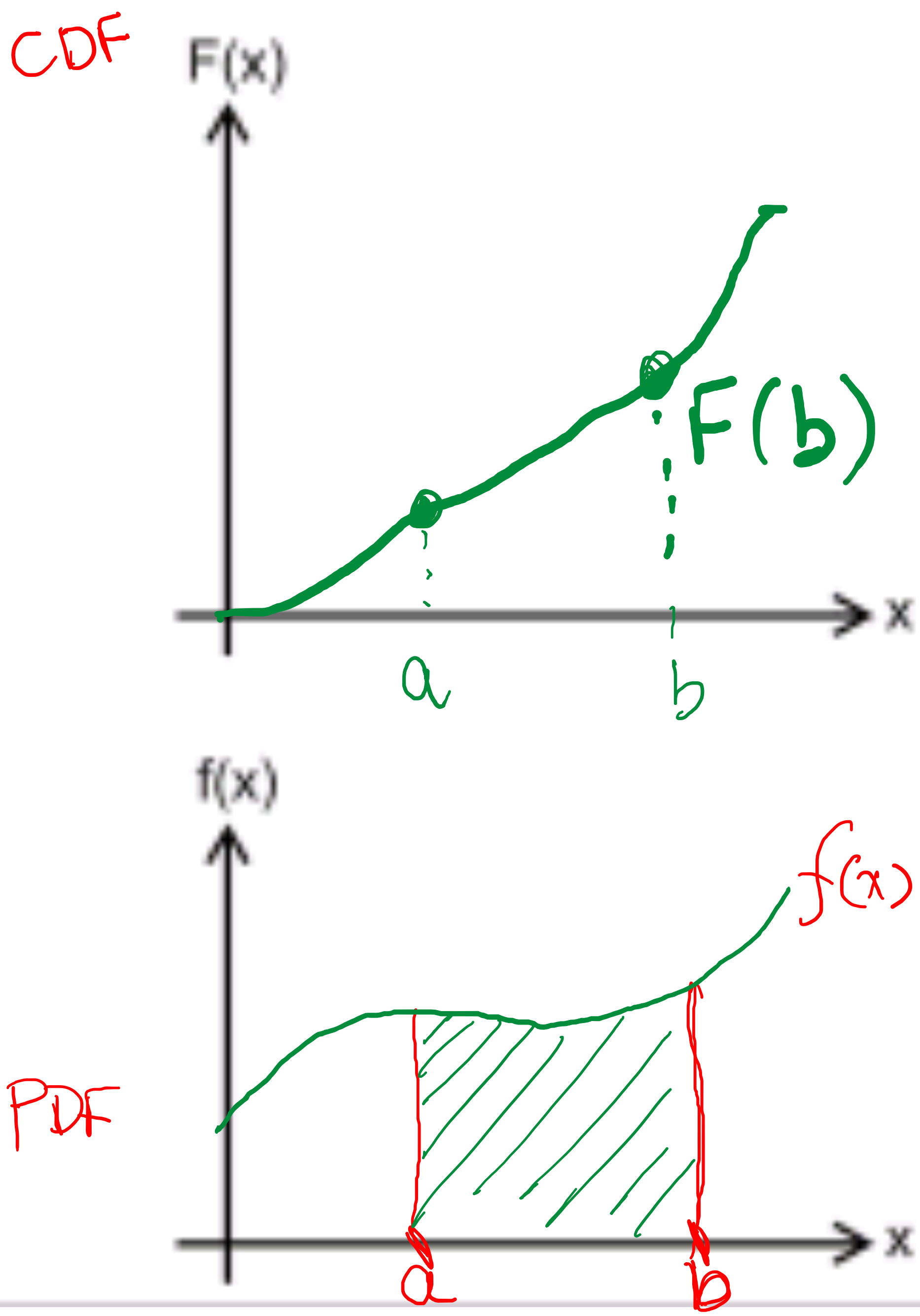
For continuous rvs, we use the derivative of the CDF F:

$$f(x) = F'(x) \quad \text{for all } x \in \mathbb{R}$$

Example: X is uniform over $[0,1]$

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \le x \le 1 \\ 1 & \text{if } x > 1 \end{cases} \qquad f(x) = \begin{cases} 1 & \text{if } 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

FUNDAMENTAL THEOREM OF CALCULUS (FTC)



$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx \iff F'(x) = f(x)$$

https://en.wikipedia.org/wiki/Fundamental_theorem_of_calculus

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CDF <-> PDF

- At the board, we recalled the Fundamental Theorem of Calculus (FTC)
- By FTC, the CDF is the antiderivative of the PDF
- This is why we refer to CDF by *F* and PDF by *f*

$$F'(x) = f(x)$$

- PDF -> CDF: integrate
- CDF -> PDF: differentiate

$$F(x) = \int_{-\infty}^{x} f(x)\, dx$$

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DISTRIBUTION FUNCTIONS

- **Example:** X is uniform over $[0,1/3]$

  We can find the CDF and PDF just like we did in our previous example when X was uniform over $[0,1]$

Note: Since X is continuous, we need to find the CDF first (compute probabilities), and get the PDF from it via differentiation

DISTRIBUTION FUNCTIONS

$$F\left(\frac{1}{6}\right) = 3 \cdot \frac{1}{6} = \frac{1}{2}$$

- Random variable: X is uniform over [0, 1/3]

$$f(x) = \begin{cases} 3 & \text{if } 0 \le x \le 1/3 \\ 0 & \text{otherwise} \end{cases}$$

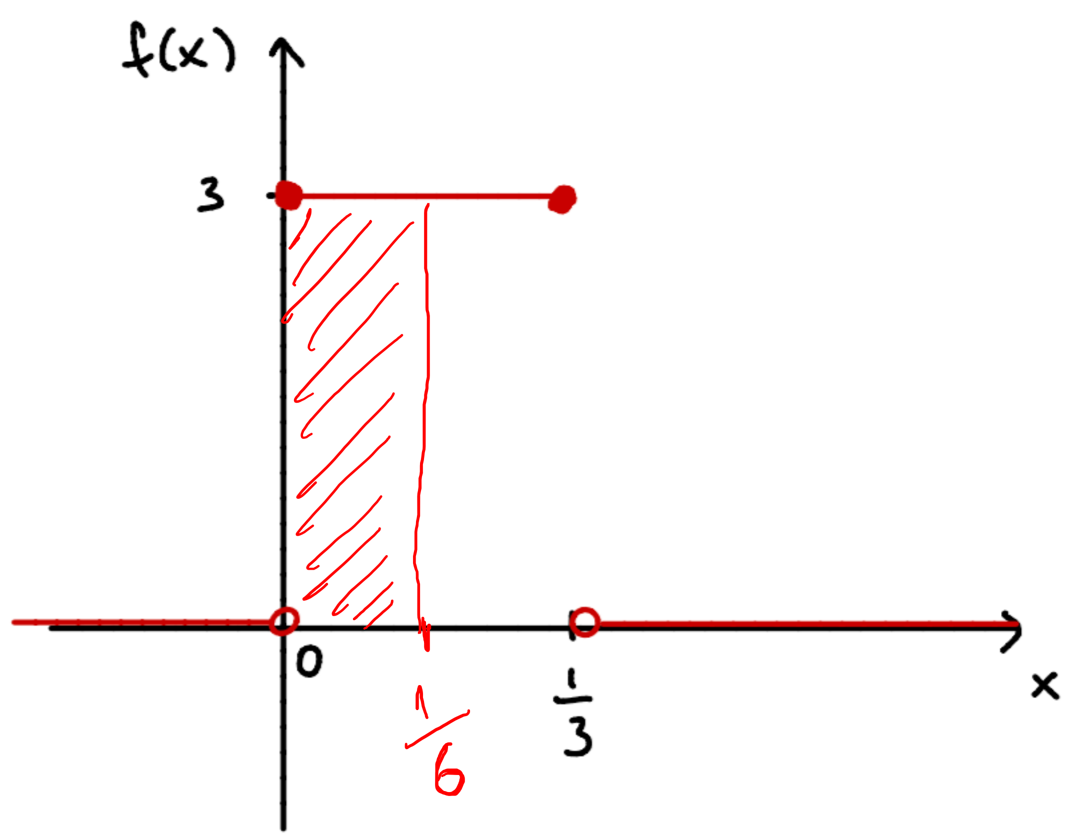
$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 3x & \text{if } 0 \le x \le 1/3 \\ 1 & \text{if } x > 1/3 \end{cases}$$

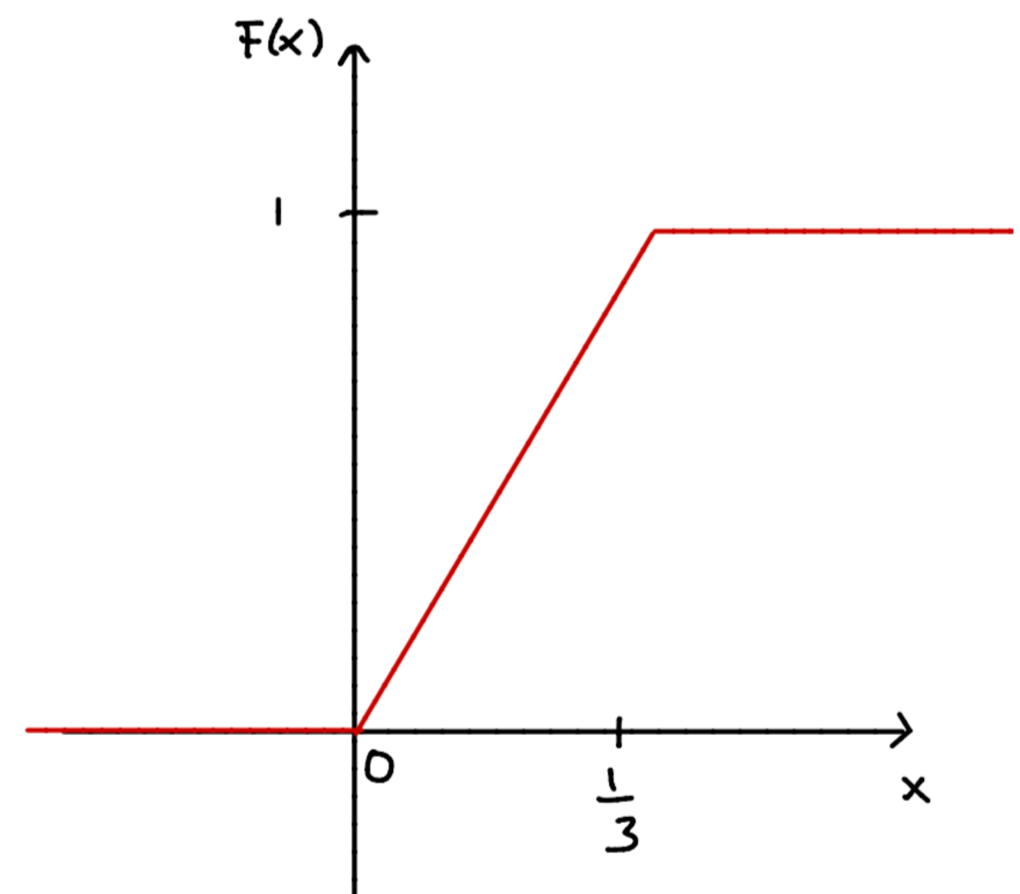
$$\Pr\left(X \le \frac{1}{6}\right) = \int_0^{1/6} 3\,dx = 3\int_0^{1/6} dx = 3\left[x\right]_0^{1/6}$$

$$= 3 \cdot \left[\tfrac{1}{6} - 0\right]$$

$$= 3 \cdot \frac{1}{6}$$

$$= \frac{1}{2}$$





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PDF: CONTINUOUS VS DISCRETE

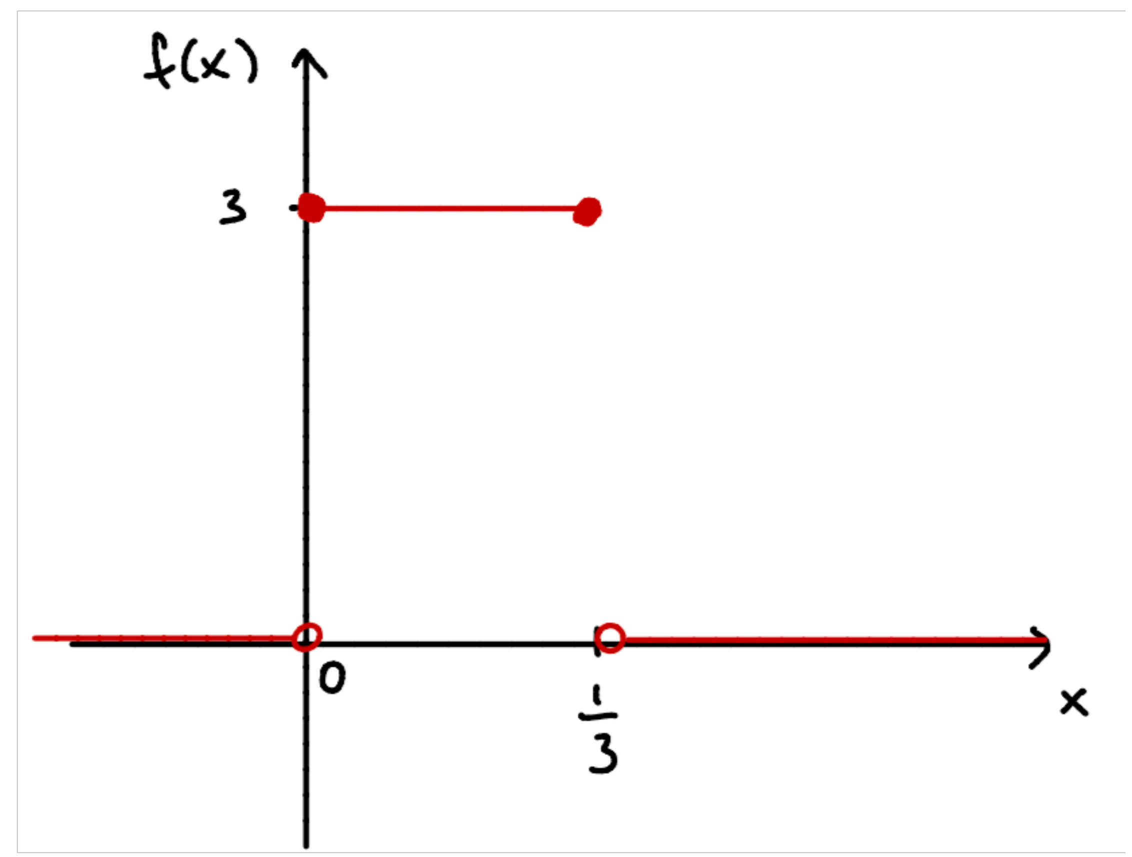
- **Example:** X is uniform over $[0,1/3]$

Note: Unlike the discrete setting, we can have $f(x) > 1$

$$f(x) = \begin{cases} 3 & \text{if } 0 \le x \le 1/3 \\ 0 & \text{otherwise} \end{cases}$$

The area under the curve needs to be 1:

$$\int_{-\infty}^{\infty} f(x)dx = \int_{0}^{1/3} 3dx = 1$$



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PROPERTIES OF THE CDF (DISCRETE AND CONTINUOUS)

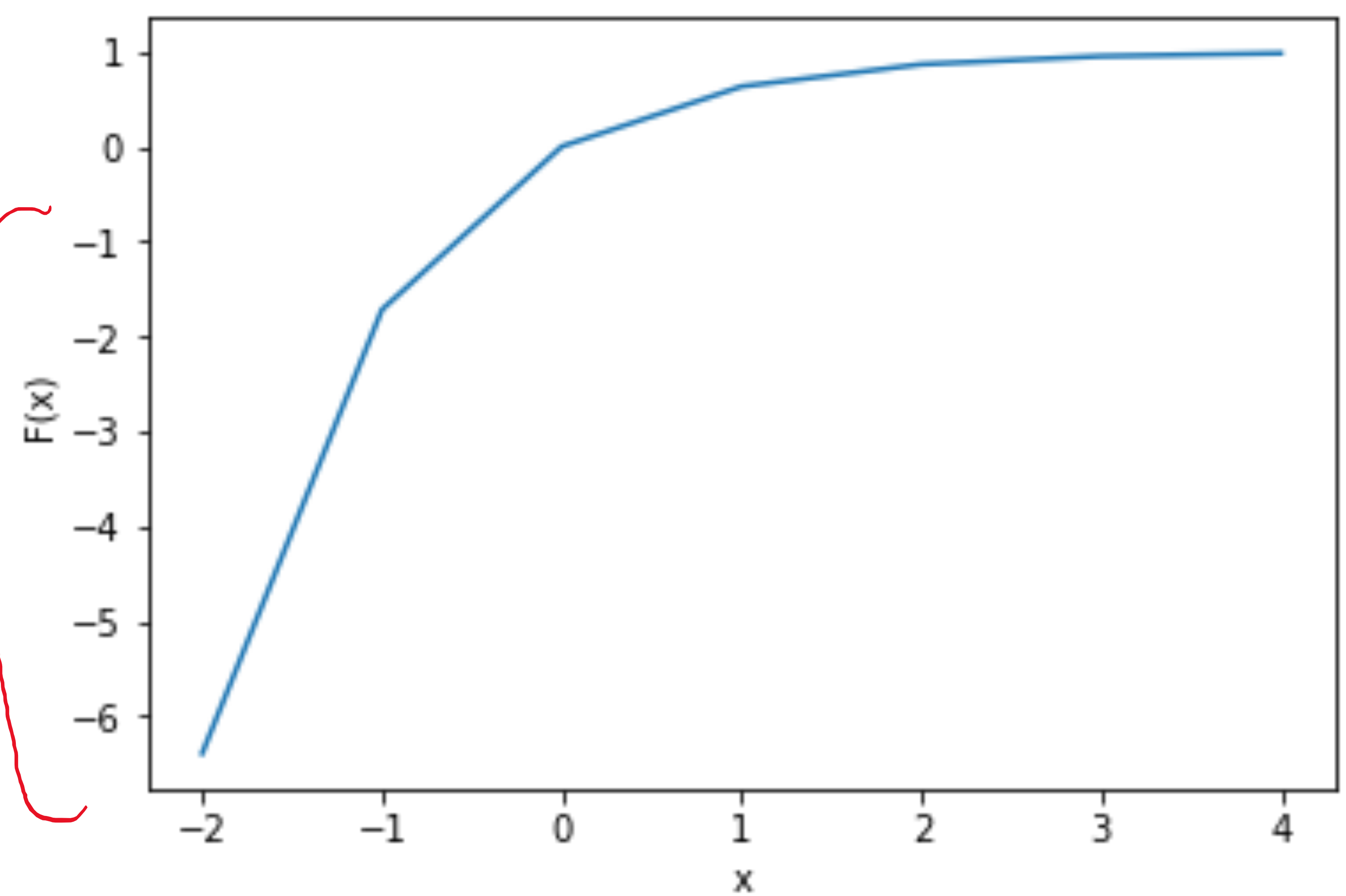
- $F(x) = Pr(X \leq x)$ (this is the definition)
- $F(x) \in [0,1]$ (it is a probability)
- $F(x)$ is nondecreasing (accumulating Pr)
- 0 to the left: $\lim_{x \to -\infty} F(x) = 0$
- 1 to the right: $\lim_{x \to +\infty} F(x) = 1$

for continuous

- $Pr(a < X \leq b) = F(b) - F(a)$
- $F'(x) = f(x)$ (by the Fundamental Theorem of Calculus)

ASK THE AUDIENCE

Can this be the plot of a valid CDF?



negative values?

(A) Yes

(B) No

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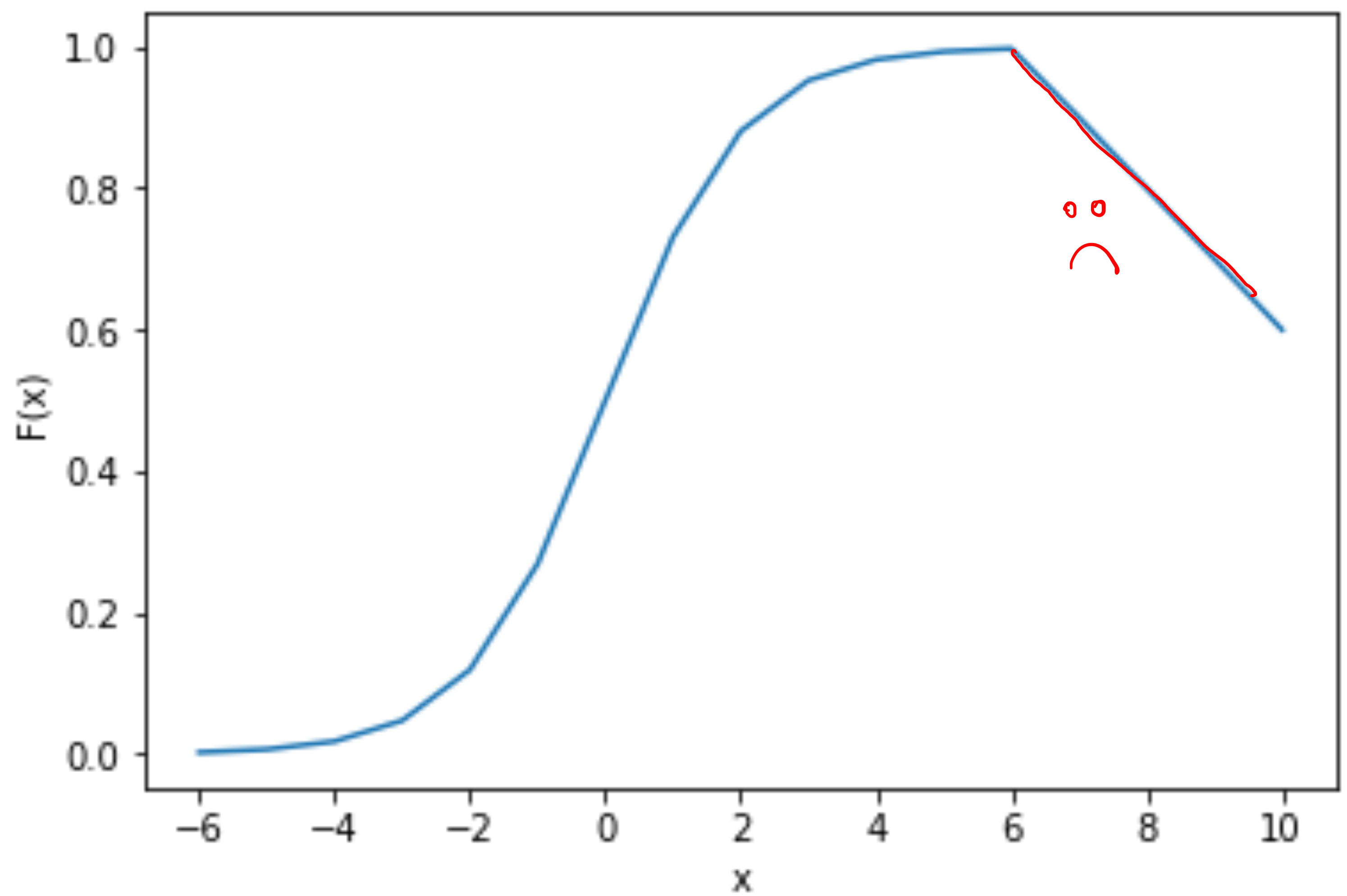
ASK THE AUDIENCE

Can this be the plot of a valid CDF?



(A) Yes

(B) No

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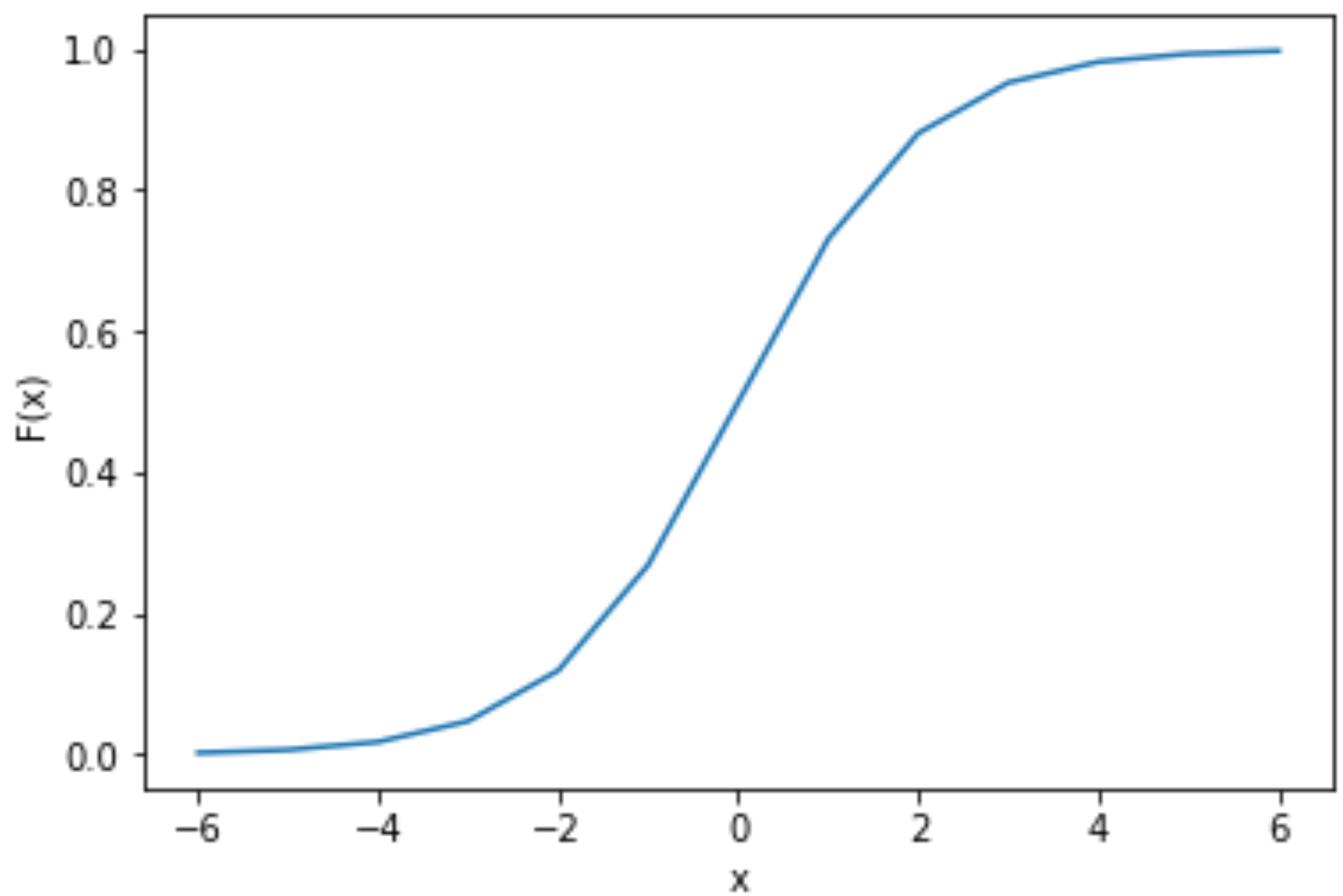
ASK THE AUDIENCE

Can this be the plot of a valid CDF?



(A) Yes

(B) No

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DISTRIBUTION FUNCTIONS

- The PDF and CDF capture the same information about the random variable
- We can obtain the PDF from the CDF and vice-versa

Consider a continuous random variable X

PDF → CDF: integrate $F(x) = \int_{-\infty}^{x} f(t)dt$

CDF → PDF: differentiate $f(x) = F'(x)$

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DISTRIBUTION FUNCTIONS

- The PMF and CDF capture the same information about the random variable
- We can obtain the PMF from the CDF and vice-versa

$\boxed{\text{Consider a discrete random variable } X}$

X has a countable set of values: $x_1 < x_2 < x_3 < \ldots$

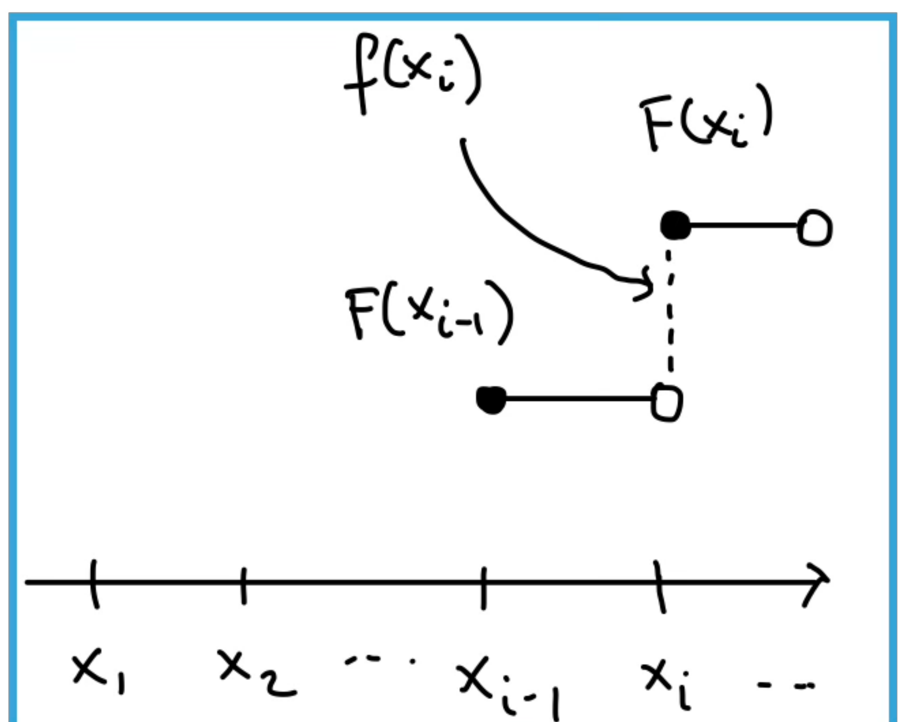
PMF → CDF: sum of PMF values

$F(x_i) = f(x_1) + \ldots + f(x_i)$ for all i

$$\Pr(X \leq x_i) = \Pr(X \in \{x_1, \ldots, x_i\}) = \Pr(X = x_1) + \ldots + \Pr(X = x_i)$$

DISTRIBUTION FUNCTIONS

- The P F and CDF capture the same information about the random variable
- We can obtain the P F from the CDF and vice-versa



CDF → P F: difference of successive CDF values

$$f(x_1) = F(x_1), f(x_i) = F(x_i) - F(x_{i-1}) \text{ if } i \geq 2$$

DISTRIBUTION FUNCTIONS

- The PMF and CDF capture the same information about the random variable

- We can obtain the PMF from the CDF and vice-versa

Consider a discrete random variable X

X has a countable set of values: $x_1 < x_2 < x_3 < \dots$

PMF → CDF: sum of P F values

$$F(x_i) = f(x_1) + \dots + f(x_i) \text{ for all } i$$

CDF → PMF: difference of successive CDF values

$$f(x_1) = F(x_1), f(x_i) = F(x_i) - F(x_{i-1}) \text{ if } i \geq 2$$